



Fig. 24: Caustic soda crystals

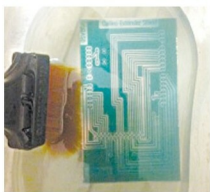


Fig. 25: PCB in developer solution

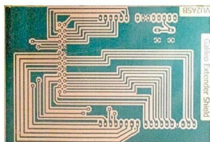


Fig. 26: Fully developed PCB

ter. The etched board would look as shown in Fig. 28.

### Step 7: Removing the masking material

Use a piece of cotton cloth dipped in nail polish remover to remove the blue-coloured toner polymer coating from the PCB.

Some nail polish removers are acetone based, and though these serve our purpose well, they are less commonly available. Many nail polish removers use a mix of ethyl acetate (solvent for nail polish) and isopropyl alcohol. This is a milder chemical but is commonly available



Fig. 27: Signs of beginning of etching

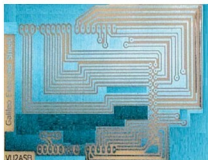


Fig. 28: Fully etched PCB

and delivers equally good results.

You may also use a paint thinner. This step will result in a PCB as shown in Fig. 29.

### Step 8: Drilling holes in the PCB

This is a critical step. The size of the holes and their precise location on the PCB are two most vital aspects you need to address. For most components, 1mm hole diameter is sufficient; some terminal blocks and diodes may need a hole diameter of 1.2mm. This aspect is not difficult to take care of. Maintaining the exact distance between holes required for multi-pin components like ICs and header blocks, though, is very crucial. The slightest misalignment of holes will make it very difficult to even push the component in its destined place.

#### Tips, pitfalls and workarounds.

I use a good-quality general-purpose PCB as a former for drilling holes on the newly etched PCB by aligning two farthest placed pins of the component pad centre. This gives excellent results. The boards have 1mm diameter holes, which are spaced

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